기타

소볼레프 공간 : $f \in H^s(\mathbb{R}^d) \iff \int_{\mathbb{R}^d} \overbrace{|\hat{f}(\xi)|^2 (1+|\xi|^2)^s}^{\text{거칠기 절두}} \, d\xi < \infty$

f 안에 ξ 성분이 얼마나 많은가?

ξ에 대한 penalty

"H^s에 가입하려면, 고주파 성분을 많이 가져서는 안된다"

- $B^s(\Omega) = \{ f : \Omega \to \mathbb{R} \ : \ \|f\|_{B^s(\Omega)} < \infty \}$

- $\|f\|_{B^s(\Omega)} := \inf_{\underbrace{f_e|\Omega = f}} \left[\int_{\mathbb{R}^d} \overbrace{(1+|\xi|)^s |\hat{f_e}|}^{\text{거칠기 절두}} \, d\xi \right]$

f를 Ω 밖으로 확장한 f_e (주리에 변환)

머신러닝의 목표 : 일반화 (generalization)

- 학습 데이터 $\{x_i, y_i\}_{i=1}^{n}$
- 손실 함수 $\ell(y, f(x))$

$$\begin{cases} \text{실제의 오류} : & E_{(Y,X)\sim P}\left[\ell(Y, f(X))\right] \\ \text{경험적 오류} : & \frac{1}{n}\sum_{i=1}^{n} \ell(Y_i, f(X_i)) \end{cases}$$

generalization error

Total error $\approx$ $\underbrace{O\left(\frac{1}{\sqrt{n}}\right)}_{\text{Estimation Error}}$ + $\underbrace{O\left(\frac{1}{m}\right)}_{\text{Approximation Error}}$

데이터 n개 유닛 m개

<Estimation Error>

- Rademacher Complexity
  - 라데마커 확률변수 $P(\sigma_i = 1) = P(\sigma_i = -1) = 50\%$ (노이즈)
  - 라데마커 복잡도 $\mathrm{Rad}_n(F) = E\left[\sup_{f\in F}\left(\frac{1}{n}\sum_{i=1}^{n} \underbrace{\sigma_i f(z_i)}_{\text{노이즈를 따라가는 정도.}}\right)\right]$ $\propto$ 과적합 위험.

---

let $F_{m,\sigma} = \left\{ f_\theta(x) = \sum_{j=1}^{m} \underbrace{\beta_j \sigma(w_j^T x)}_{NN} \right\}$

then $\mathrm{Rad}(F_{m,\sigma}, \theta) \lesssim \frac{\theta}{\sqrt{n}}$

θ ← 파라미터의 크기 제한

n ← 데이터의 샘플 갯수

UAT의 의의 : NN은 $C(I_n)$에서 dense하다.

.. 한계 : NN의 width는 얼마여야 하는가?

curse of dimensionality!

· L^p norm : $\|f\|_{L^p} = \left(\int_X |f(x)|^p \, d\mu\right)^{1/p}$

$= \left(\int_\Omega |f(x)|^p \, dx\right)^{1/p}$ where $\Omega \subseteq \mathbb{R}^d$

보통 Ω.

· 푸리에 변환 : $\hat{f}(\xi) = \int_{\mathbb{R}^d} f(x) \cdot e^{-2\pi i \langle \xi, x \rangle} \, dx$

· 역푸리에 변환 : $f(x) = \int_{\mathbb{R}^d} \hat{f}(\xi) \, e^{2\pi i \langle \xi, x \rangle} \, d\xi$

· 바론 클래스 : $\mathcal{P}_{C_1} = \{ f \in L^1(\mathbb{R}^d) : \|\hat{f}\|_{L^1(\mathbb{R}^d)} < \infty$

and $\int_{\mathbb{R}^d} |2\pi\xi| \, |\hat{f}(\xi)| \, d\xi < C_1 \}$

$L^1(\mathbb{R}^d)$

: 함수의 절대값을 $\mathbb{R}^d$에 대해 적분했을 때 유한한 값을 가진다.

· 2계층 신경망 : $\Phi(x) = \sum_{i=1}^{N} a_i \, \sigma(w_i^T x_i + b_i)$

정리 sigmoid σ,

$f \in \Gamma_{c_1}$, $c > 4c_1^2$, $N \in \mathbb{N}$

B_1^d : 반지름 1짜리 공 $\subseteq \mathbb{R}^d$

then 有 Φ_{NN} s.t $\frac{1}{|B_1^d|}\int_{B_1^d} |f(x) - \Phi(x)|^2 \, dx \le \frac{c}{N}$ ← d와는 무관하다!

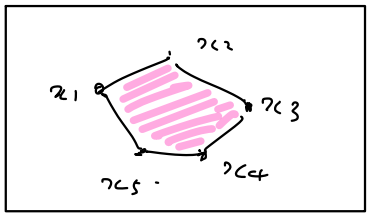
보조정리 ①

- H : Hilbert Space

$$co(G) = \left\{ \sum_{j=1}^{n} \alpha_j x_j : \begin{array}{l} n \in \mathbb{N},\ x_j \in G, \\ \alpha_j > 0,\ \sum_{j=1}^{n} \alpha_j = 1 \end{array} \right\}$$

- $G \subseteq H$: 어떤 $B > 0$에 대해 모든 $g \in G$가 $\|g\|_H \le B$인 집합

$\|g\|_H \le B$



- $f \in \overline{co}(G)$ (closure of the convex hull of G)

- $c > B^2$

- then 有 $(g_i)_{i=1\dots}^{N} \subseteq G$ s.t $\left\| f - \frac{1}{N}\sum_{i=1}^{N} g_i \right\|_H^2 \le \frac{c}{N}$

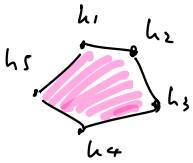
<증명 1 | 증명>

Step 1

- since $f \in \overline{co(G)}$
- if f is limit point, $\forall \varepsilon > 0$

有 $f^* \in co(G)$ s.t $\|f - f^*\|_H < \varepsilon$

$$f^* = \sum_{j=1}^{m} \alpha_j \cdot h_j$$



$\alpha_j > 0$ $\sum \alpha_j = 1$, $h_i \in G$

Step 2 claim

$\exists \{\theta_1, \dots, \theta_N\} \subseteq \{h_1, \dots, h_m\}$ s.t $\left\| f^* - \frac{1}{N} \sum_{j=1}^{N} \theta_j \right\|_H^2 \le \frac{B^2}{N}$

Step 3 let $P(X_i = h_j) = \alpha_j$

then $E[X_i] = \sum_{j=1}^{m} P(X_i = h_j) \cdot h_j$

$= \sum_{j=1}^{m} \alpha_j h_j = f^*$

Step 4

- $E\left[\| f^* - \frac{1}{N} \sum_{i=1}^{N} X_i \|^2\right]$

$$= \frac{1}{N^2} E\left[\| \sum_{i=1}^{N} (f^* - X_i) \|^2\right]$$

$$= \frac{1}{N^2} E\left[\sum_{i=1}^{N} \| f^* - X_i \|^2 + \sum_{i \neq j} \langle \cancel{f^* - X_i}, \cancel{f^* - X_j} \rangle\right]$$

$\begin{cases} X_i, X_j \text{는 독립} \\ E[X_i] = f^* \end{cases}$

$$= \frac{1}{N^2} E\left[\sum_{i=1}^{N} \| f^* - X_i \|^2\right]$$

$$= \frac{1}{N^2} \cdot N \cdot E\left[\cancel{\|f^*\|^2} - 2 \cdot \langle \cancel{f^*}, X_i \rangle + \|X_i\|^2\right] = \frac{1}{N}\left(E[\|X_i\|^2] - \|f^*\|^2\right)$$

$$\le \frac{B^2}{N}$$

- $E[\|X_i\|^2] = \sum_{j=1}^{m} P(X_i = h_j) \cdot \|h_j\|^2$

$= \sum_{j=1}^{m} \alpha_j \|h_j\|^2$

$\le \sum_{j=1}^{m} \alpha_j B^2 \le B^2$

<정리 증명>

$$G_{C_1} = \{ x \mapsto \mathbb{1}_{\mathbb{R}_+}(\langle a, x\rangle + b) \ : \ a \in \mathbb{R}^d, \ b \in \mathbb{R}, \ \|a\| \le 2C_1 \}$$

$\begin{cases} +1 & : \text{양수} \\ 0 & : \text{음수} \end{cases}$

Lemma 2. $f \in \mathcal{P}_{C_1}$ 이면, $f(x) - f(0) \in \overline{\mathrm{co}(G_{C_1})}$ 이다.

$$f(x) - f(0) = \int_{\mathbb{R}^d} \hat{f}(\xi)\left(e^{2\pi i \langle \xi, x\rangle} - 1\right) d\xi$$

$$= \int_{\mathbb{R}^d} |\hat{f}(\xi)| \left\{ \cos[2\pi\langle x, \xi\rangle + k(\xi)] - \cos k(\xi) \right\} d\xi \qquad (\text{실수부})$$

"무한히 많은 코사인의 가중합"

- $f \in \mathcal{P}_{C_1}$ 이면 $f(x) - f(0) \in \overline{\mathrm{co}}(G_{C_1})$ 이다.

- $\begin{cases} H : L^2(B_1^d) \\ G : G_{C_1} \\ \text{유계} : \|g\|_{L^2} \le 2C_1 \\ \quad (g \in G_{C_1}) \end{cases}$

보조정리 ①

$\Downarrow$

$\sigma(x)$

- $$\frac{1}{|B_1^d|} \int_{B_1^d} \left| (f(x) - f(0)) - \boxed{\sum_{i=1}^{N} w_i \mathbb{1}_{\mathbb{R}^+}(\cancel{\langle a_i, x\rangle} + b_i)} \right|^2 dx \le \frac{(2C_1)^2}{N} + \delta$$

$w_i \sigma(\langle \tilde{a}_i, x\rangle + \tilde{b}_i)$

- $\lim_{r \to \infty} \sigma(rx) = \mathbb{1}_{\mathbb{R}^+}(x)$ (아주 가파른 시그모이드)

<Estimation Error>

· Rademacher Complexity

· 라데마커 확률변수 $P(\xi_i = 1) = P(\xi_i = -1) = 50\%$ (노이즈)

· 라데마커 복잡도 $\mathrm{Rad}_n(F) = E\left[\sup_{f \in F}\left(\frac{1}{n}\sum_{i=1}^{n} \underbrace{\xi_i f(z_i)}_{\text{노이즈를 따라가는 정도.}}\right)\right] \propto$ 과적합 위험.

---

let $F_{m,\sigma} = \left\{ f_\theta(x) = \sum_{j=1}^{m} \underbrace{\beta_j \sigma(w_j^T x)}_{NN} \right\}$

then $\sup_{f \in F_{m,\sigma}} \left| \underbrace{\frac{1}{n}\sum_{i=1}^{n} f(x_i)}_{\text{empirical risk}} - \underbrace{E[f]}_{\text{population risk}} \right| \le 2\,\mathrm{Rad}(F_{m,\sigma}) + \sqrt{\frac{1}{2n} \cdot \log\frac{2}{\delta}}$ with (prob) $\ge 1-\delta$

also, $\mathrm{Rad}(F_{m,\sigma,Q}) \le \frac{Q}{\sqrt{n}}$ $\quad F_{m,\sigma,Q} = \{ f \in F_{m,\sigma} : \|f\| \le Q \}$

(유기)

$B > 0$ 이고, 레이블기 $\|z_i\|_2 \le C$ 이라

σ가 ReLU 이라면 $\Rightarrow \quad Rad(F_{m,\sigma,B}) \le \dfrac{2BC}{\sqrt{n}}$

- 2계층 신경망 $\quad f_\theta^{(m)} = \sum_{j=1}^{m} \beta_j \sigma(w_i^T x)$

- 복잡도 척도 $\quad C(\theta) = \sum_{j=1}^{m} |\beta_j| \|w_j\|_2$

- $F_{m,\sigma,B} = \{ f_\theta \in F_{m,\sigma} : C(\theta) \le B \}$

Positive Homogeneity

- ReLU의 객실 성질 $\quad \alpha \sigma(x) = \sigma(\alpha x) \quad \forall \alpha > 0$

- 라라메터 재조정 $\quad \theta = \{(\beta_j, w_j)\}_{j \le m}$

$\theta' = \{(\lambda_j \beta_j, w_j / \lambda_i)\}_{j \le m} \quad (\lambda_i > 0)$

- 가중치나 방향의 분리 $\quad$ let $\lambda_j = \|w_j\|_2$

$\sigma(w_j^T x) = \|w_j\|_2 \cdot \sigma(\bar{w}_j^T x)$ → 방향 단위벡터

- 라데마커 복잡도 추측

$$\begin{cases} F_1 = \{ x \mapsto w^T x : w \in \mathbb{R}^d, \ \|w\|_2 \le 1 \} \\ f_\theta(x) = \sum \beta_j \sigma(w_j^T x) = \sum \beta_j \|w_j\|_2 \sigma(\bar{w}_j^T x) \end{cases}$$

- $Rad(F_{m,\sigma,B}) \le 2B \cdot Rad(F_1) = 2B \cdot \left(\frac{C}{\sqrt{n}} \right)$

Note. 달라스랑의 축약 원리

$Rad(\phi \circ F) \le K \cdot Rad(F)$

when ϕ : 립시츠 함수

K : 립시츠 상수

ReLU는 립시츠 함수이다.